IY3660 – Applications of Cryptography

Week 2 – Lecture
Module overview, scene setting and classic ciphers

Dr Elizabeth Quaglia
Information Security Group

ROYAL HOLLOWAY UNIVERSITY OF LONDON

In today’s lecture

- Module overview
- Scene setting
- Classic ciphers

Module overview

Who

- The course will be run by **Elizabeth Quaglia**, supported by **Benjamin Bencina** and **Simon Pohmann**
  - Elizabeth will deliver the lectures and run the workshops
  - Benjamin and Simon will support the labs and give some guest lectures

Where, when, how

Every week we will have scheduled teaching sessions on

- Wednesday, 10:00 to 11:50, in Bourne Annexe (lecture)
- Thursday, 11:00 to 11:50, in BEDFORD-0-04-06 (workshop or **lab**)

In some weeks, we will run a **lab session** on Thursdays.

<u>Office hours</u>: Wednesday 13:00-14:00, or by appointment

Labs

- Labs in weeks 4, 5, 6, 8, 9, 12 (six labs in total)
  - First lab session: 16 Oct (11:00 in BEDFORD-0-04-06)
- Labs will reinforce your learning in class
- We will be using Python/Sage for running the labs

Moodle & MS Teams

- **Moodle**: will host assignments, course material, further reading
- **MS Teams**: used when running course activities online (if necessary)
- Announcements will be posted on Moodle

Summative assessment

There will be two forms of summative (for credit) assessment:

- Written **exam** next May (70%).
- **One coursework** this semester (30%).
  - Due date: **Thursday 4th December at 2pm**
  - Released two weeks ahead of the due date
  - May consist of challenging questions (requiring independence of thought, and maybe some research) or a short essay
  - Will not involve programming

Quizzes and research activity

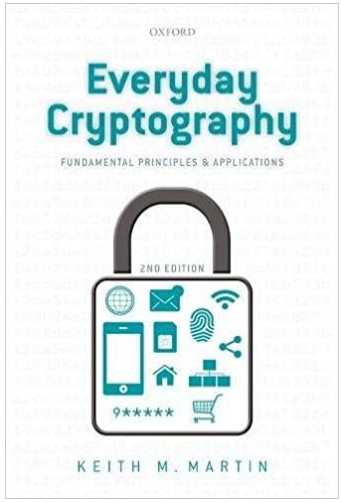
- We will also have quizzes (short questions) at the beginning of lectures
- Occasionally, I might assign self-study lectures
- Suggested research activity ("Further Reading & Research" sections) is available on Moodle
- These activities are designed to deepen your understanding of concepts covered in class

On the use of GenAI

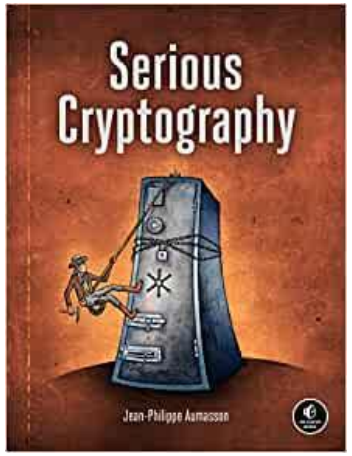
- If used it needs to be declared and properly referenced (see College guidelines)
- Its output cannot be cut & pasted – you need to elaborate on it. Otherwise this will be considered *commission*, a **major** offence.

Recommended books

- **Keith Martin, *Everyday Cryptography* (2nd edition)**
  - Oxford University Press 2017, ISBN: 0198788010
  - Broad coverage, mostly non-mathematical
  - Recommended purchase



- **Jean Philippe Aumasson, *Serious Cryptography***
  - No Starch Press, 2017, ISBN 978-1593278267
  - a “practical guide” to modern cryptography
  - Recommended purchase



Additional resources to support your learning

- **Applied Cryptography Knowledge Area – CyBOK 1.1.0,** by K. G. Paterson
  - "provides a broad introduction to the field of cryptography, focusing on applied aspects of the subject"
  - It features essentially the same structure as our course. A highly recommended reading!

- **A Graduate Course in Applied Cryptography,** by D. Boneh and V. Shoup
  - comprehensive and in-depth coverage of applied cryptography – goes deeper and beyond what we cover in the course
  - excellent resource to support your learning and for students who wish to learn more about applied cryptography

Technical prerequisites: we assume that…

- You have taken a previous module in Information Security
  - But you may not remember anything about cryptography from that module
- You know and care about about bits, bytes, logical operators,…
- You know about complexity and algorithms
- You remember A-level mathematics
  - But we'll be teaching you some more
- You can already program proficiently in Python
- You already know how to use the command line at a basic level

Your attitude to learning is important

- We assume you want to be here and be intellectually challenged
- We assume you will come to classes and labs **on time**, ready to learn
- We assume you will ask questions if you don’t understand something
    - We will ask questions to test your understanding
- We assume you will tell us if some aspect of the course is not working for you
    - Please do not save it until the feedback questionnaire at the very end – the sooner we know, the sooner we can address it!

Objectives of the course

- The aim of this module is to explain the fundamentals behind cryptography and how it is deployed in real-world systems
  - The security services that can be provided by cryptography
  - The main cryptographic mechanisms, their security goals, design and basic attacks
  - Real-world applications and deployment of cryptography

- We're going to give you a broad overview of the key topics in cryptography that you are likely to run into as a practitioner (e.g., software developer).
  - Aim is to prevent you from making embarrassing and costly mistakes later

Objectives of the course

- The selected approach to present this material: focus on how to use cryptography to enable secure communications
  - We will see how to use and combine the several cryptographic tools to set up a channel that can be used to securely exchange data between two communicating parties

- Note: this is an introductory course, and we will barely scratch the surface
  - You will not be designing your own cryptographic algorithms!
  - We will provide pointers to enable you to dig deeper

Learning outcomes

By the end of this module, you should be able to:

- Explain what cryptography can be used for
- Describe several cryptographic mechanisms for providing each of the core security services
- Appreciate the differences between various types of cryptographic mechanisms and in which situations they are most usefully employed
- Identify the issues that need to be addressed when assessing what types of cryptographic mechanism are necessary to secure an application
- Identify the limitations of cryptography and how to support it within a full security architecture

Module overview

- **Scene setting:** motivation, historical background
- **Symmetric cryptography**: stream ciphers, block ciphers, modes of operation, hash functions, password hashing, message authentication codes (MACs), Authenticated Encryption
- **Asymmetric cryptography**: public-key encryption, digital signatures, post-quantum cryptography, PKI, key management
- **Cryptographic protocols**: the TLS protocol (bringing it all together...)

Questions?

Over to you

Let’s go to vevox!

Scene setting

Definitions

- **Cryptography**: the art of secret writing
- Derived from the Greek:
  - *kryptos* (meaning ``hidden'')
  - *grafo* (meaning ``write’’)

- A bit of terminology:
  - **Cryptography**: “hidden” writing
  - **Cryptanalysis**: “breaking” ciphers
  - **Cryptology**: Cryptography + Cryptanalysis
  - **Steganography**: “covered” writing

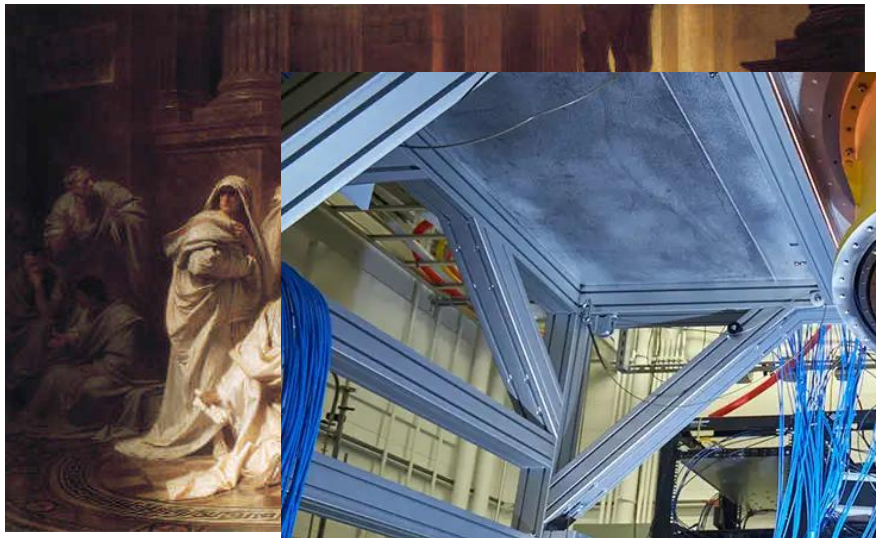
From history to the modern age

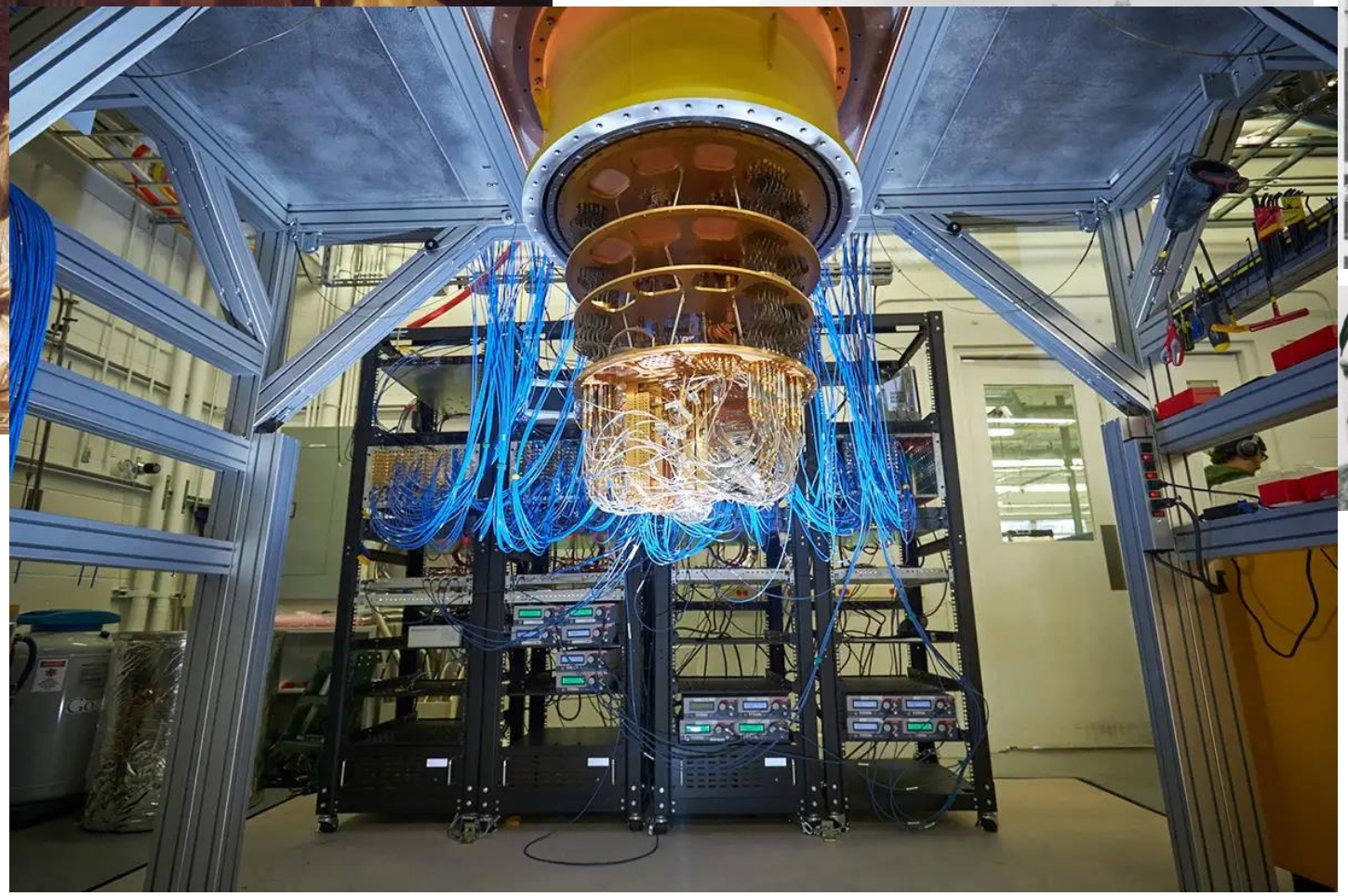
- Cryptography has been used for centuries by parties wishing to communicate securely
- Historically, it is associated with *encryption* (to provide confidentiality)
- Cryptography has now developed into a significant area at the intersection between computer science, mathematics and systems engineering, and plays a crucial role in information security

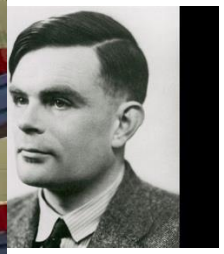
Modern cryptography

- No longer limited to confidentiality, no longer limited to communications
- Cryptography can be used to provide
  - *confidentiality*
  - *integrity* and *authenticity*
  - *authentication*
- Cryptography can be used to
  - *restrict access* to data
  - secure data at rest
  - enable some interesting modern applications
    - E.g., compute a function *securely*

As our world changes, so does cryptography







A modern definition of cryptography

From the *Handbook of Applied Cryptography* (`http://cacr.uwaterloo.ca/hac`):

Cryptography is the study of mathematical techniques related to aspects of information security such as confidentiality, data integrity, entity authentication, and data origin authentication

- Definition covers cryptographic **design**: how to **achieve** certain security goals
  - Modern approach: prove that desired goals are indeed achieved

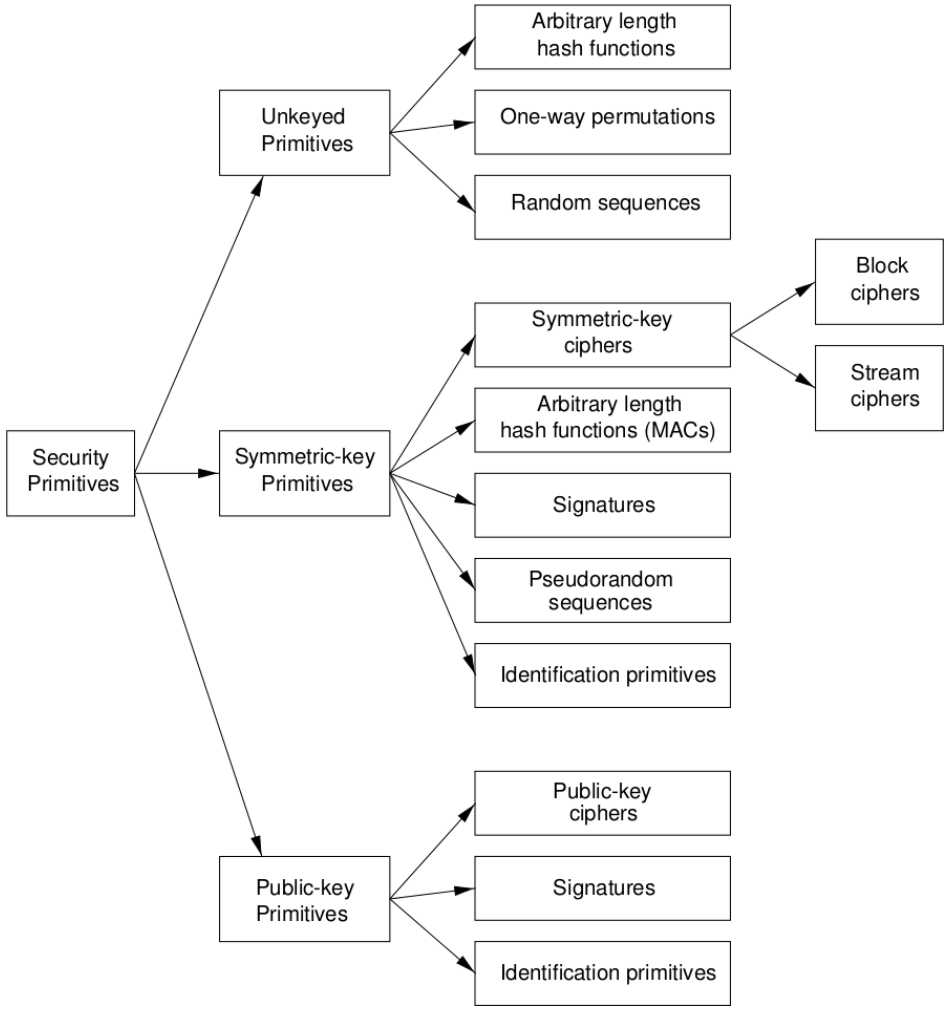
- Definition covers **cryptanalysis**: how to **break** certain security goals
  - System is 'broken' if it is weaker than expected/claimed
  - Note: not required to break it completely (e.g., recover used keys)
    - But rather show that not all security goals have been achieved!

Glossary of security services

Confidentiality	The assurance that data cannot be viewed by an unauthorised entity (sometimes referred to as secrecy)
Integrity	The assurance that data has not been altered in an unauthorised (which includes accidental) manner since the time that the data was last created, transmitted or stored by an authorised entity
Entity authentication	The assurance that a given entity is involved and currently active in a communication session (sometimes referred to as identification)
Data origin authentication	The assurance that a given entity was the original source of some data (sometimes referred to as message authentication)
Non-repudiation	Preventing the denial of previous commitments or actions

Cryptographic primitives

- More or less all cryptographic systems used in applications today are made up of basic components, or **primitives**
- These are combined to provide **security services** from the previous slide
- We will discuss most of them in this module

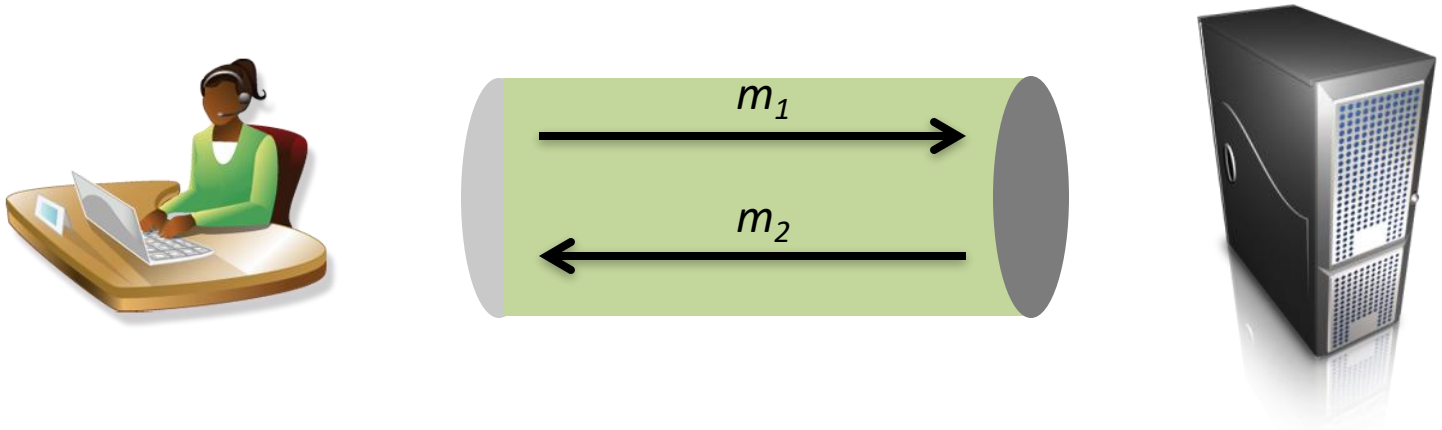


© Handbook of Applied Cryptography

The importance of security infrastructure

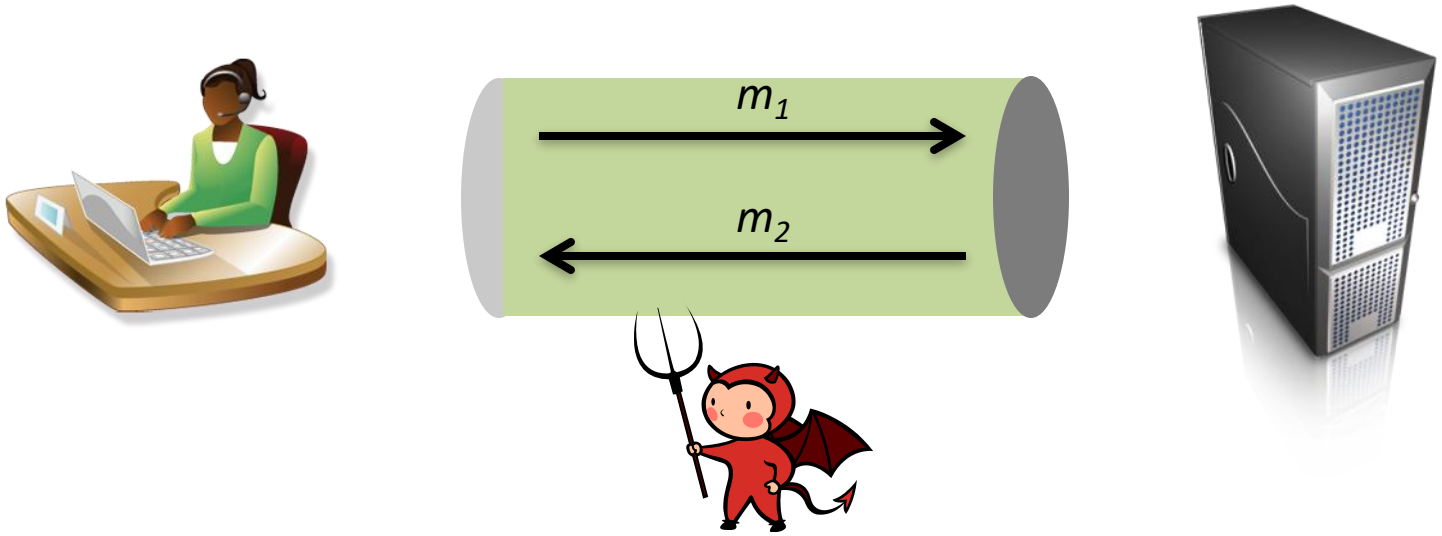
- A fundamental principle is that a **security infrastructure** must be put in place to support the deployment of any cryptographic solution.
- By **infrastructure** we mean the procedures, plans, policies, and management to make sure that any deployment actually does the job for which it was intended
  - Cryptography on its own is not a magic bullet
  - This is a principle worth keeping in mind throughout the module and more generally, throughout your studies and career

Our main application: secure communications



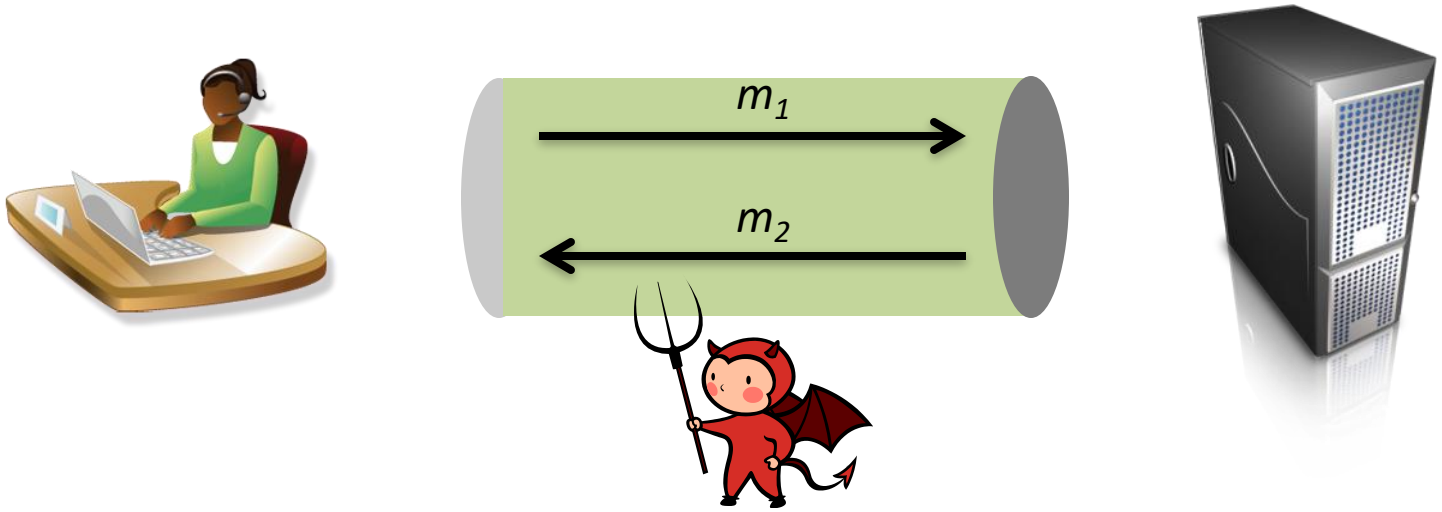
- Two entities exchange a sequence of messages over an insecure *channel*
- Entities are often called *Alice* and *Bob*, but they need not be people
- The insecure channel will be provided by a communications network, e.g., a wireless LAN, a mobile phone network, the Internet, or a combination of these
- Alice and Bob use cryptography to build a secure communications channel on top of the insecure channel

Our main application: secure communications



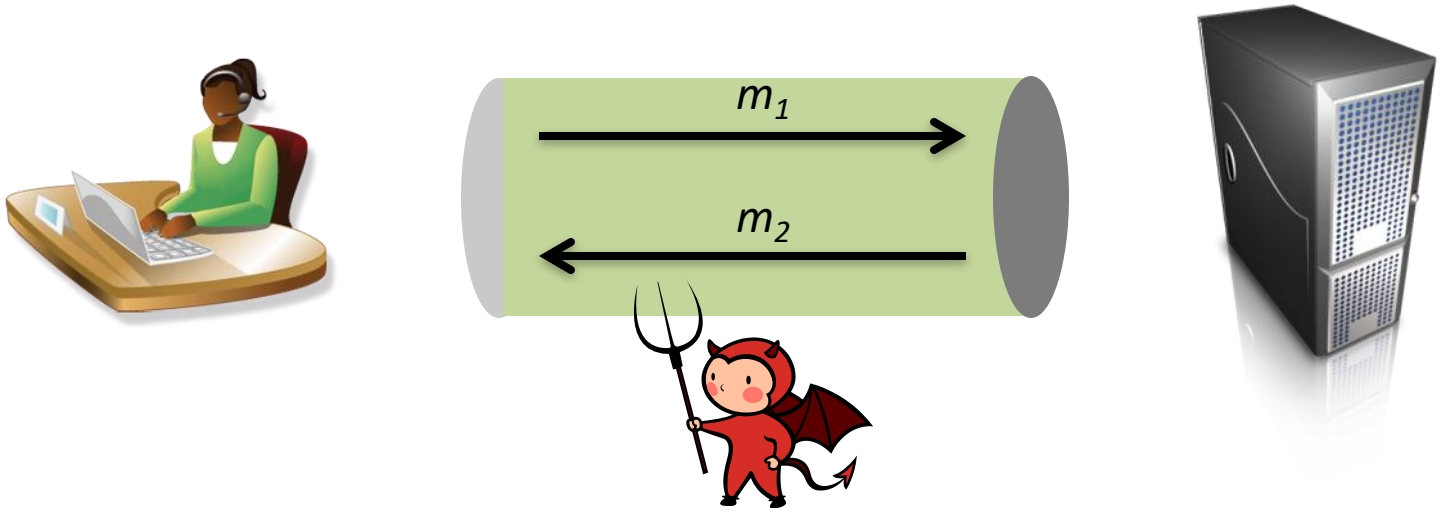
- What should our security goals be?
- What capabilities does the adversary have?
- How can we use cryptography to achieve our goals in the face of this adversary?

Secure communications: security goals



- The messages that are exchanged should remain **confidential**
- It should be hard for the adversary to modify the messages without being detected – Alice and Bob should be able to verify the **integrity** of the messages
- It should be hard for the adversary to inject messages of its own into the sequence – Alice and Bob should be able to check the **origin** of the messages
- Alice and Bob should be able to detect when messages are deleted
- Alice and Bob should be able to detect when messages are reordered (possibly by the adversary; or by the network)

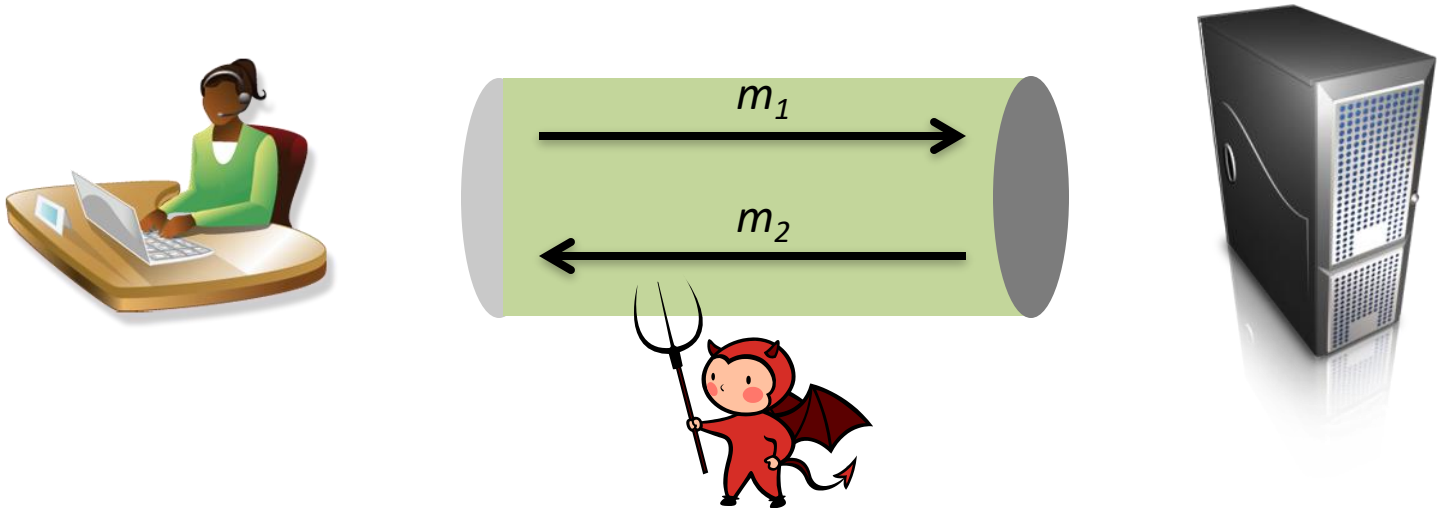
Secure communications: adversarial capabilities



- The adversary can see all the data being transferred on the network (passive adversary)
- The adversary has sufficient control over the network that it can delete, delay, modify and reorder network packets at will (active adversary)

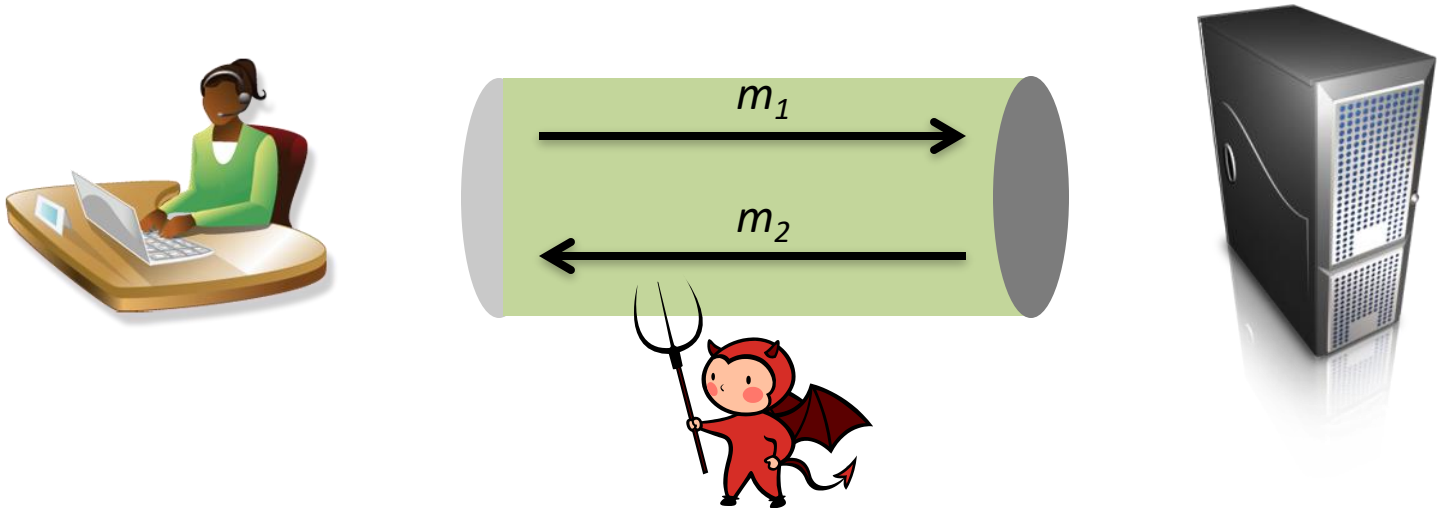
Q: To what extent are these capabilities realistic?

Secure communications: adversarial capabilities



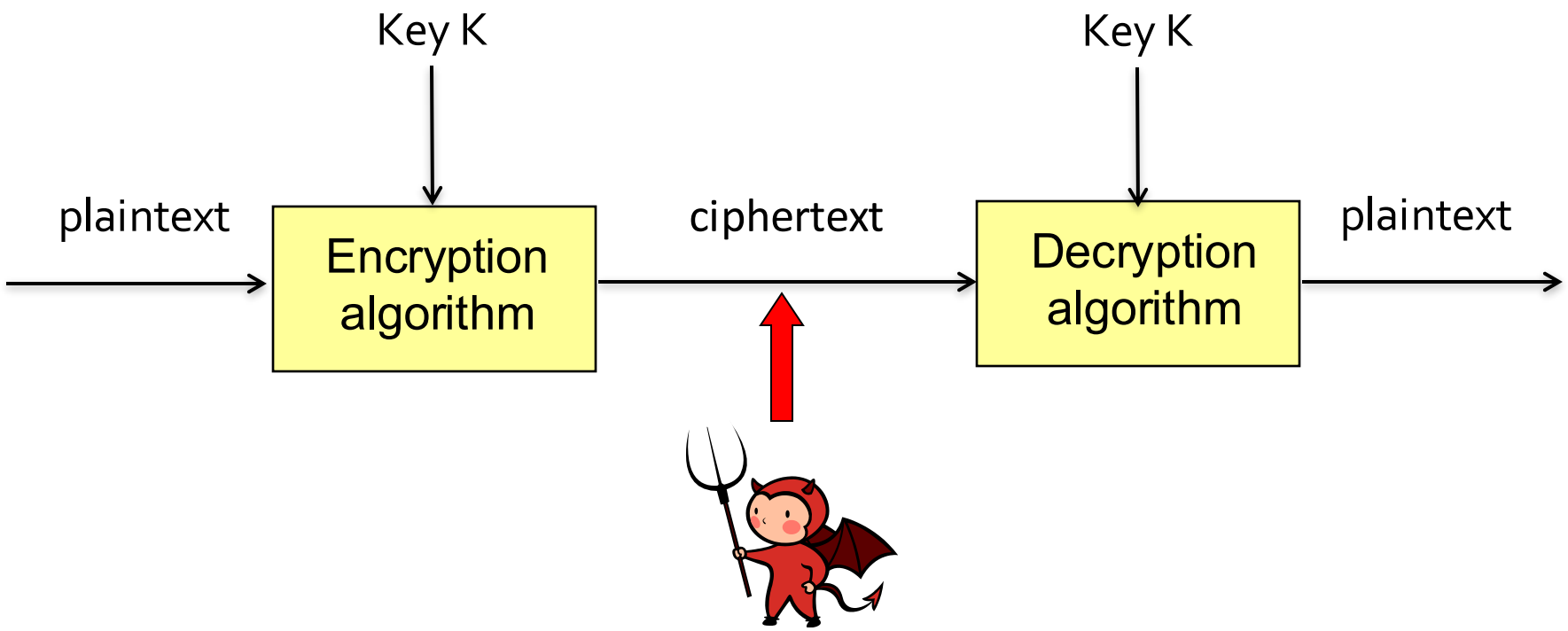
- The adversary can ask for specific messages *m* of its choice to be passed over the secure channel (**chosen plaintext attack**)
- The adversary can observe the effects on network messages when it injects network packets of its own (**chosen ciphertext attack**)

Secure communications: our objective



Our objective is to learn how to use cryptography to build secure channels which protect communicating parties against powerful adversaries

Secure communications: a simplified model



- In symmetric key cryptography, the encryption key and decryption key are the same
- We then refer to the *secret key*, or just "the key", usually denoted K
- Later we'll look asymmetric systems, where encryption and decryption keys are *not* the same

Kerckhoffs' principles

List of requirements on cryptosystems articulated by Kerckhoffs in 1883:

1° Le système doit être matériellement, sinon mathématiquement, indéchiffrable ;
2° Il faut qu'il n'exige pas le secret, et qu'il puisse sans inconvénient tomber entre les mains de l'ennemi ;
3° La clef doit pouvoir en être communiquée et retenue sans le secours de notes écrites, et être changée ou modifiée au gré des correspondants ;
4° Il faut qu'il soit applicable à la correspondance télégraphique ;
5° Il faut qu'il soit portatif, et que son maniement ou son fonctionnement n'exige pas le concours de plusieurs personnes ;
6° Enfin, il est nécessaire, vu les circonstances qui en commandent l'application, que le système soit d'un usage facile, ne demandant ni tension d'esprit, ni la connaissance d'une longue série de règles à observer.

- System should be, if not theoretically unbreakable, unbreakable in practice
- Compromise of system details should not inconvenience the correspondents
- Key should be memorable without notes and easily changed
- Cryptogram (ciphertext) should be transmissible by telegraph
- Encryption apparatus should be portable, operable by a single person
- System should be easy, requiring neither the knowledge of a long list of rules nor mental strain

Kerchoffs' principles

Modern interpretation: The security of a cryptosystem should only rely on the (secrecy of) keys used. It should <u>not depend</u> on the secrecy of the system details

Wrong interpretation: No security through obscurity

Right interpretation: Obscurity can help improve security, but systems can and will be reverse-engineered. The system should still be secure even when its specification is completely known. Furthermore, open specification may encourage review and analysis, improving our confidence in the system's security

Use of cryptography: individual perspective

- Cryptography makes our lives more convenient, enabling us to shop online, interact with our government's services online, work from home, etc
- It also enables established human rights such as privacy and freedom of expression in an on-line world
- It can be argued that, in a free society, individuals should have the right to use cryptography in any way they see fit
- Indeed, a stated commitment to this freedom is one indicator of a truly free society

Use of cryptography: business perspective

- Cryptography, through the provision of security services, enables new forms of business to be carried out, allows costs to be reduced, and ensures regulatory compliance for existing forms of business
- Cryptography may also bring new costs to a business
- Moving business online may introduce new threats which cryptography can only partially address, or which it does not address at all
- Cryptography will only be deployed if it makes business sense – cost-effective, appropriate, compliant with regulations

Use of cryptography: government perspective

- Governments have conflicting requirements with respect to cryptography:
- On the one hand they may wish to promote a competitive and attractive business environment, with cryptography being an enabler. They may also wish to streamline their operations by moving services on-line
- On the other hand, governments may wish to control crime and manage issues of national security by limiting the use of cryptography. This could be via the imposition of laws and regulations, by the promotion of weak cryptographic standards, or by other means

UK government perspective

Our vision for 2021 is that the UK is secure and resilient to cyber threats, prosperous and confident in the digital world.

The UK Cyber Security Strategy (2016)

Our vision is that the UK in 2030 will continue to be a leading responsible and democratic cyber power, able to protect and promote our interests in and through cyberspace in support of national goals.

The UK Cyber Security Strategy (2022)

https://www.gov.uk/government/publications/national-cyber-strategy-2022/national-cyber-security-strategy-2022

Access to cryptography

- The past decade has seen the widening of discussion about rights to privacy-protection cryptographic mechanisms

Search **Bloomberg** Sign In

Technology

Facebook, WhatsApp Will Have to Share Messages With U.K.

By Kitty Donaldson and Mark Burton
28 September 2019, 11:46 BST *Updated on 28 September 2019, 22:23 BST*

- Information will be shared in serious criminal investigations
- Facebook opposes gove

LIVE ON BLOOMBERG

The New York Times

Australian Government Passes Contentious Encryption Law

The law, opposed by privacy advocates, requires tech companies to provide law enforcement and security agencies with access to encrypted communications.

Home / Technology / Internet
Home / Technology / Security

JULY 24, 2019

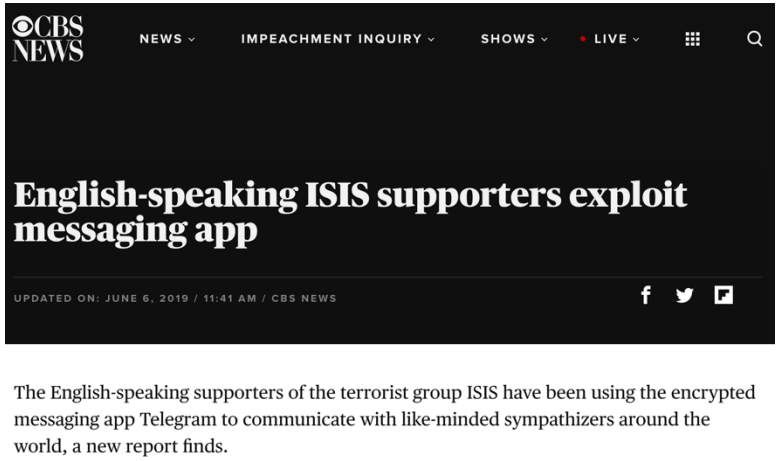
Barr warns time's running out for companies to open encryption

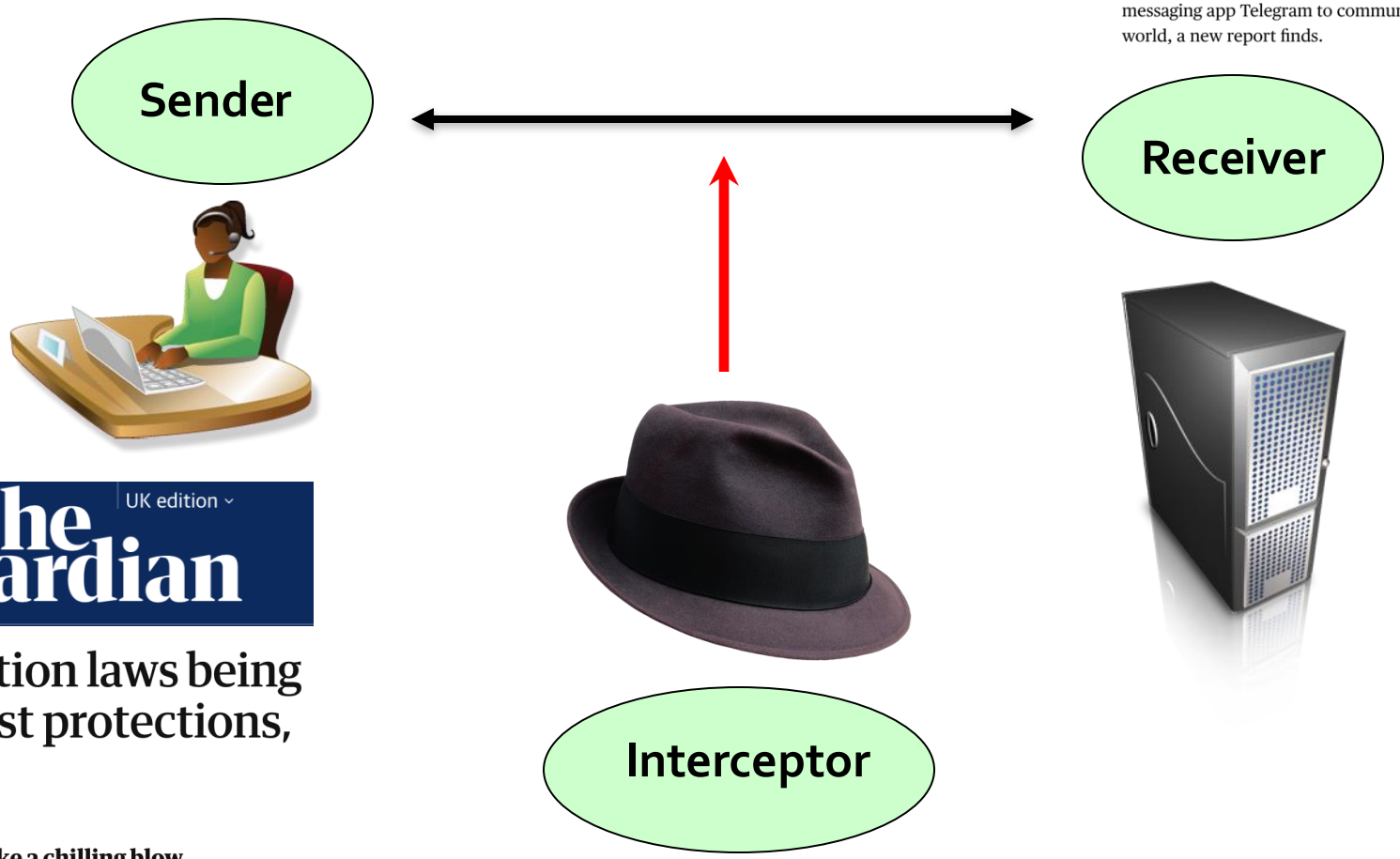
by Chris Strohm

Credit: CC0 Public Domain

Attorney General William Barr issued a sharp warning that time may be running out for Facebook Inc. and other technology companies to come to a voluntary agreement providing law enforcement officials with access to the encrypted communications of their users.

Access to cryptography





Questions?

Classic ciphers

Some historical ciphers

- Transposition Cipher
  - Skytale

- Substitution Cipher
  - Monoalphabetic Ciphers
    - Caesar Cipher
  - Polyalphabetic Ciphers
    - Vigenère Cipher

Transposition cipher

- If you have a message with *n* characters, just shuffle the characters to produce the ciphertext.
  - Mathematically, consider a permutation of *n* characters
  - There are *n!* permutations
- Example of Transposition Cipher
  - Skytale – the earliest apparatus used in cryptography (by the Spartans in the 5th century BC)

What is the key?

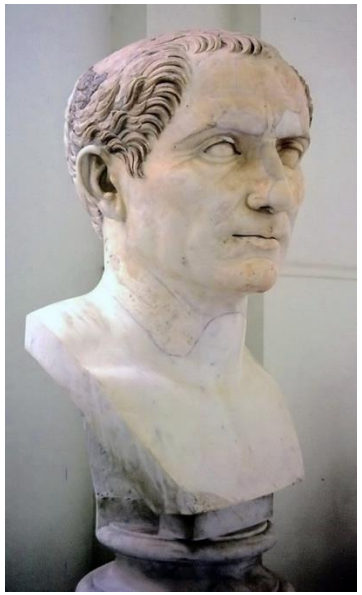
Substitution cipher - Caesar cipher

- Simple algorithm, supposedly used by Julius Caesar.
  - Classic version: during encryption, characters are shifted 3 positions forward:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Enc ↓ ↑ *Dec*

  - For decryption, characters are shifted 3 positions *backwards*.
  - This idea can be generalised to K positions (0,…, 25):
    - E.g, if K = 7, then A -> H, B -> I, C -> J , etc.
    - The secret key is the shift K.
    - Mathematically, $\mathcal{E}(c, k) = (c + k) \ (mod\ 26)$



Example of encrypted text

TY QZFC TYLOPBFLEP HZCOD LWLY EFCTYR LAAPLCD YZH LD ESP QZFYOPC ZQ NZXAFEPC DNTPYNP, ESP ZCTRTYLEZC ZQ ESP OZXTYLYE EPNSYZWZRJ ZQ ESP WLEP EHPYETPES NPYEFCJ, MFE ESPDP HZCOD HPCP YZE DAZVPY TY STD ZHY WTQPETXP, LYO SP XLJ JPE MP DPPY TY L OTQQPCPYE WTRSE TY ESP QFEFCP.

How would you decipher it?

Example of encrypted text

TY QZFC TYLOPBFLEP HZCOD LWLY EFCTYR LAAPLCD YZH LD ESP QZFYOPC ZQ NZXAFEPC DNTPYNP, ESP ZCTRTYLEZC ZQ ESP OZXTYLYE EPNSYZWZRJ ZQ ESP WLEP EHPYETPES NPYEFCJ, MFE ESPDP HZCOD HPCP YZE DAZVPY TY STD ZHY WTQPETXP, LYO SP XLJ JPE MP DPPY TY L OTQQPCPYE WTRSE TY ESP QFEFCP.

We could try all the keys: $K = 0, \ldots, K = 25$

Example of encrypted text

TY QZFC TYLOPBFLEP HZCOD LWLY EFCTYR LAAPLCD YZH LD ESP QZFYOPC ZQ NZXAFEPC DNTPYNP, ESP ZCTRTYLEZC ZQ ESP OZXTYLYE EPNSYZWZRJ ZQ ESP WLEP EHPYETPES NPYEFCJ, MFE ESPDP HZCOD HPCP YZE DAZVPY TY STD ZHY WTQPETXP, LYO SP XLJ JPE MP DPPY TY L OTQQPCPYE WTRSE TY ESP QFEFCP.

Try $K = 3$:

QV NWCZ QVILMYCIBM EWZLA ITIV BCZQVO IXXMIZA VWE IA BPM NWCVLMZ WN KWUXCBMZ AKQMVKM, BPM WZQOQVIBWZ WN BPM LWUQVIVB BMKPVWTWOG WN BPM TIBM BEMVBQMBP KMVBCZG, JCB BPMAM EWZLA EMZM VWB AXWSMV QV PQA WEV TQNMBQUM, IVL PM UIG GMB JM AMMV QV I LQNNMZMVB TQOPB QV BPM NCBCZM.

Example of encrypted text

TY QZFC TYLOPBFLEP HZCOD LWLY EFCTYR LAAPLCD YZH LD ESP QZFYOPC ZQ NZXAFEPC DNTPYNP, ESP ZCTRTYLEZC ZQ ESP OZXTYLYE EPNSYZWZRJ ZQ ESP WLEP EHPYETPES NPYEFCJ, MFE ESPDP HZCOD HPCP YZE DAZVPY TY STD ZHY WTQPETXP, LYO SP XLJ JPE MP DPPY TY L OTQQPCPYE WTRSE TY ESP QFEFCP.

Try $K = 7$:

MR JSYV MREHIUYEXI ASVHW EPER XYVMRK ETTIEVW RSA EW XLI JSYRHIV SJ GSQTYXIV WGMIRGI, XLI SVMKMREXSV SJ XLI HSQMRERX XIGLRSPSKC SJ XLI PEXI XAIRXMIXL GIRXYVC, FYX XLIWI ASVHW AIVI RSX WTSOIR MR LMW SAR PMJIXMQI, ERH LI QEC CIX FI WIIR MR E HMJJIVIRX PMKLX MR XLI JYXYVI.

Example of encrypted text

TY QZFC TYLOPBFLEP HZCOD LWLY EFCTYR LAAPLCD YZH LD ESP QZFYOPC ZQ NZXAFEPC DNTPYNP, ESP ZCTRTYLEZC ZQ ESP OZXTYLYE EPNSYZWZRJ ZQ ESP WLEP EHPYETPES NPYEFCJ, MFE ESPDP HZCOD HPCP YZE DAZVPY TY STD ZHY WTQPETXP, LYO SP XLJ JPE MP DPPY TY L OTQQPCPYE WTRSE TY ESP QFEFCP.

Try $K = 11$:

IN FOUR INADEQUATE WORDS ALAN TURING APPEARS NOW AS THE FOUNDER OF COMPUTER SCIENCE, THE ORIGINATOR OF THE DOMINANT TECHNOLOGY OF THE LATE TWENTIETH CENTURY, BUT THESE WORDS WERE NOT SPOKEN IN HIS OWN LIFETIME, AND HE MAY YET BE SEEN IN A DIFFERENT LIGHT IN THE FUTURE.

Security of the Caesar cipher

The Caesar cipher is obviously just too weak:

- If the adversary knows that the Caesar cipher is being used, it only needs to try (up to) 26 different keys, until the text makes sense…
- **Lesson**: one of the requirements for a secure cipher is that the **key space** has to be large enough to preclude **exhaustive key search** in practice

Monoalphabetic ciphers

- The Caesar Cipher is an example of a **Monoalphabetic Cipher**
- Instead of a simple shift by *k* positions, we could use a more complex substitution…

$$\mathcal{E}(x,(a,b)) = a * x + b \ mod \ 26$$

or more generally, any permutation of 26 characters

Enc ↓

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
G	Z	A	L	T	N	X	C	R	W	D	S	B	F	Y	U	E	I	Q	V	H	M	O	J	P	K

↑ *Dec*

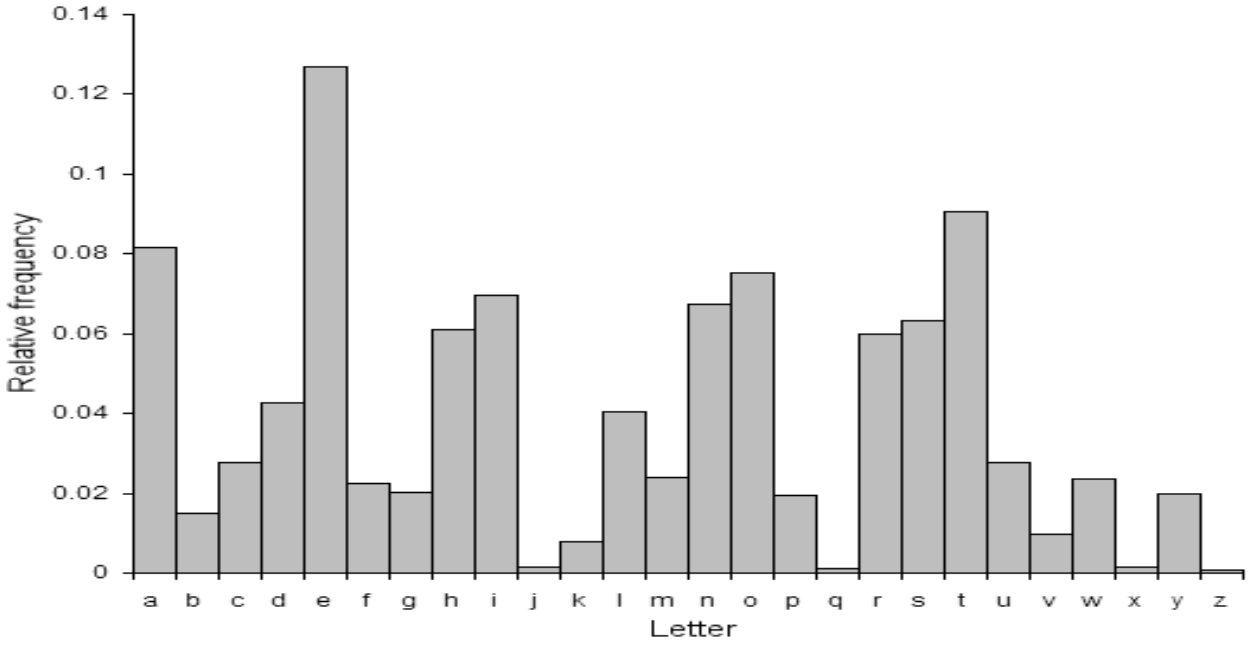
- We now have 26! = 26.25.24…2.1 ~ 10^{27} possible keys!!
  - **Exhaustive search** (trying all keys) should not be possible…

Cryptanalysis of monoalphabetic ciphers

VHIRFX OYIDTL RF RQYSGVRYF NIYB VCT UYOTINHS QACYYS YN SYXRAGS VCTYIP ATFVITL YF ACHIAC GV UIRFATVYF HFRMTIQRVP, GFL CRQ OYID TBTIXTL GQ VCGV YN G AYBUSTVT YHVQRLTI. YFT AGF YFSP QUTAHSGVT, ZHV RV SYYDQ GQ RN VHIRFX NYHFL RF VCT AYFATUV YN VCT VHIRFX BGACRFT QYBTVCRFX VCGV OYHSL QGVRQNP VCT NGQARFGVRYF ORVC VCT UIYZSTB YN BRFL VCGV ACIRQVYUCTI BYIAYB CGL QUGIDTL; CRQ VYVGS YIRXRFGSRVP SGP RF QTTRFX VCT ITSTMGFAT YN BGVCTBGVRAGS SYXRA VY G UIYZSTB YIRXRFGSSP QTTF GQ YFT YN UCPQRAQ. RF VCRQ UGUTI, GQ RF QY BGFP GQUTAVQ YN CRQ SRNT, CT BGLT G ZIRLXT ZTVOTTF VCT SYXRAGS GFL VCT UCPQRAGS OYISLQ, VCYHXCV GFL GAVRYF, OCRAC AIYQQTL AYFMTFVRYFGS ZYHFLGIRTQ.

Language statistics

- In a typical text, some characters are much more frequent than the others
  - Example: character frequency in English



Cryptanalysis of monoalphabetic ciphers

VHIRFX OYIDTL RF RQYSGVRYF NIYB VCT UYOTINHS QACYYS YN SYXRAGS VCTYIP ATFVITL YF ACHIAC GV UIRFATVYF HFRMTIQRVP, GFL CRQ OYID TBTIXTL GQ VCGV YN G AYBUSTVT YHVQRLTI. YFT AGF YFSP QUTAHSGVT, ZHV RV SYYDQ GQ RN VHIRFX NYHFL RF VCT AYFATUV YN VCT VHIRFX BGACRFT QYBTVCRFX VCGV OYHSL QGVRQNP VCT NGQARFGVRYF ORVC VCT UIYZSTB YN BRFL VCGV ACIRQVYUCTI BYIAYB CGL QUGIDTL; CRQ VYVGS YIRXRFGSRVP SGP RF QTTRFX VCT ITSTMGFAT YN BGVCTBGVRAGS SYXRA VY G UIYZSTB YIRXRFGSSP QTTF GQ YFT YN UCPQRAQ. RF VCRQ UGUTI, GQ RF QY BGFP GQUTAVQ YN CRQ SRNT, CT BGLT G ZIRLXT ZTVOTTF VCT SYXRAGS GFL VCT UCPQRAGS OYISLQ, VCYHXCV GFL GAVRYF, OCRAC AIYQQTL AYFMTFVRYFGS ZYHFLGIRTQ.

- We obtain the statistics of the ciphertext

Cryptanalysis of monoalphabetic ciphers

VHIRFX OYIDTL RF RQYSGVRYF NIYB VCT UYOTINHS QACYYS YN SYXRAGS VCTYIP ATFVITL YF ACHIAC GV UIRFATVYF HFRMTIQRVP, GFL CRQ OYID TBTIXTL GQ VCGV YN G AYBUSTVT YHVQRLTI. YFT AGF YFSP QUTAHSGVT, ZHV RV SYYDQ GQ RN VHIRFX NYHFL RF VCT AYFATUV YN VCT VHIRFX BGACRFT QYBTVCRFX VCGV OYHSL QGVRQNP VCT NGQARFGVRYF ORVC VCT UIYZSTB YN BRFL VCGV ACIRQVYUCTI BYIAYB CGL QUGIDTL; CRQ VYVGS YIRXRFGSRVP SGP RF QTTRFX VCT ITSTMGFAT YN BGVCTBGVRAGS SYXRA VY G UIYZSTB YIRXRFGSSP QTTF GQ YFT YN UCPQRAQ. RF VCRQ UGUTI, GQ RF QY BGFP GQUTAVQ YN CRQ SRNT, CT BGLT G ZIRLXT ZTVOTTF VCT SYXRAGS GFL VCT UCPQRAGS OYISLQ, VCYHXCV GFL GAVRYF, OCRAC AIYQQTL AYFMTFVRYFGS ZYHFLGIRTQ.

- We obtain the statistics of the ciphertext
  - T is the most frequent character (52 times)
    - **T** most probably corresponds to **E**

Cryptanalysis of monoalphabetic ciphers

VHIRFX OYIDTL RF RQYSGVRYF NIYB VCT UYOTINHS QACYYS YN SYXRAGS VCTYIP ATFVITL YF ACHIAC GV UIRFATVYF HFRMTIQRVP, GFL CRQ OYID TBTIXTL GQ VCGV YN G AYBUSTVT YHVQRLTI. YFT AGF YFSP QUTAHSGVT, ZHV RV SYYDQ GQ RN VHIRFX NYHFL RF VCT AYFATUV YN VCT VHIRFX BGACRFT QYBTVCRFX VCGV OYHSL QGVRQNP VCT NGQARFGVRYF ORVC VCT UIYZSTB YN BRFL VCGV ACIRQVYUCTI BYIAYB CGL QUGIDTL; CRQ VYVGS YIRXRFGSRVP SGP RF QTTRFX VCT ITSTMGFAT YN BGVCTBGVRAGS SYXRA VY G UIYZSTB YIRXRFGSSP QTTF GQ YFT YN UCPQRAQ. RF VCRQ UGUTI, GQ RF QY BGFP GQUTAVQ YN CRQ SRNT, CT BGLT G ZIRLXT ZTVOTTF VCT SYXRAGS GFL VCT UCPQRAGS OYISLQ, VCYHXCV GFL GAVRYF, OCRAC AIYQQTL AYFMTFVRYFGS ZYHFLGIRTQ.

- We obtain the statistics of the ciphertext
  - Other frequent characters are **F** (40), **G** (41), **R** (46), **V** (47), **Y** (50)
    - These probably correspond to **A**, **R**, **I**, **O**, **T**, **N**

Cryptanalysis of monoalphabetic ciphers

VHIRFX OYIDTL RF RQYSGVRYF NIYB **TCE** UYOTINHS QACYYS YN SYXRAGS **TCEORP** ATFVITL YF ACHIAC GV UIRFATVYF HFRMTIQRVP, GFL CRQ OYID TBTIXTL GQ VCGV YN G AYBUSTVT YHVQRLTI. YFT AGF YFSP QUTAHSGVT, ZHV RV SYYDQ GQ RN VHIRFX NYHFL RF **TCE** AYFATUV YN **TCE** VHIRFX BGACRFT QYBTVCRFX VCGV OYHSL QGVRQNP **TCE** NGQARFGVRYF ORVC **TCE** UIYZSTB YN BRFL VCGV ACIRQVYUCTI BYIAYB CGL QUGIDTL; CRQ VYVGS YIRXRFGSRVP SGP RF Q**EEIN**X **TCE** ITSTMGFAT YN BG**TCE**BGVRAGS SYXRA VY G UIYZSTB YIRXRFGSSP QTTF GQ YFT YN UCPQRAQ. RF VCRQ UGUTI, GQ RF QY BGFP GQUTAVQ YN CRQ SRNT, CT BGLT G ZIRLXT ZTVOTTF **TCE** SYXRAGS GFL **TCE** UCPQRAGS OYISLQ, VCYHXCV GFL GAVRYF, OCRAC AIYQQTL AYFMTFVRYFGS ZYHFLGIRTQ.

- We obtain the statistics of the ciphertext
  - Other frequent characters are **F** (40), **G** (41), **R** (46), **V** (47), **Y** (50).
    - These probably correspond to **A**, **R**, **I**, **O**, **T**, **N**
    - Now **C** should correspond to **H,** while **P** corresponds to **Y**

Cryptanalysis of monoalphabetic ciphers

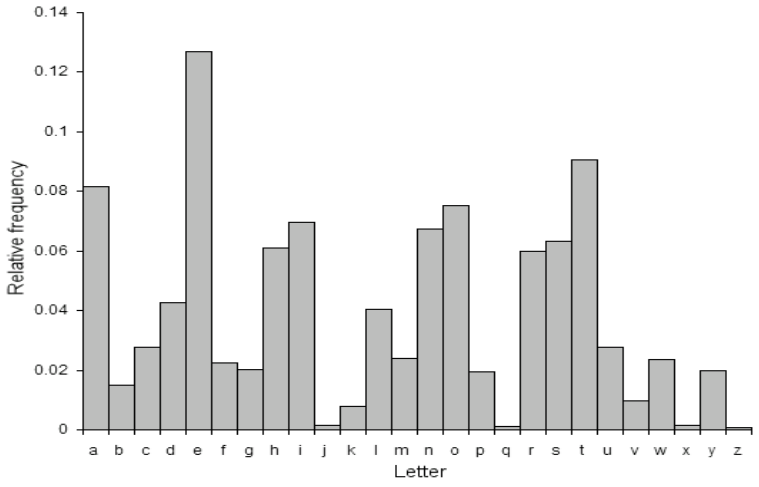
- We can continue this process until we obtain:

TURING WORKED IN ISOLATION FROM THE POWERFUL SCHOOL OF LOGICAL THEORY CENTRED ON CHURCH AT PRINCETON UNIVERSITY, AND HIS WORK EMERGED AS THAT OF A COMPLETE OUTSIDER. ONE CAN ONLY SPECULATE, BUT IT LOOKS AS IF TURING FOUND IN THE CONCEPT OF THE TURING MACHINE SOMETHING THAT WOULD SATISFY THE FASCINATION WITH THE PROBLEM OF MIND THAT CHRISTOPHER MORCOM HAD SPARKED; HIS TOTAL ORIGINALITY LAY IN SEEING THE RELEVANCE OF MATHEMATICAL LOGIC TO A PROBLEM ORIGINALLY SEEN AS ONE OF PHYSICS. IN THIS PAPER, AS IN SO MANY ASPECTS OF HIS LIFE, HE MADE A BRIDGE BETWEEN THE LOGICAL AND THE PHYSICAL WORLDS, THOUGHT AND ACTION, WHICH CROSSED CONVENTIONAL BOUNDARIES.

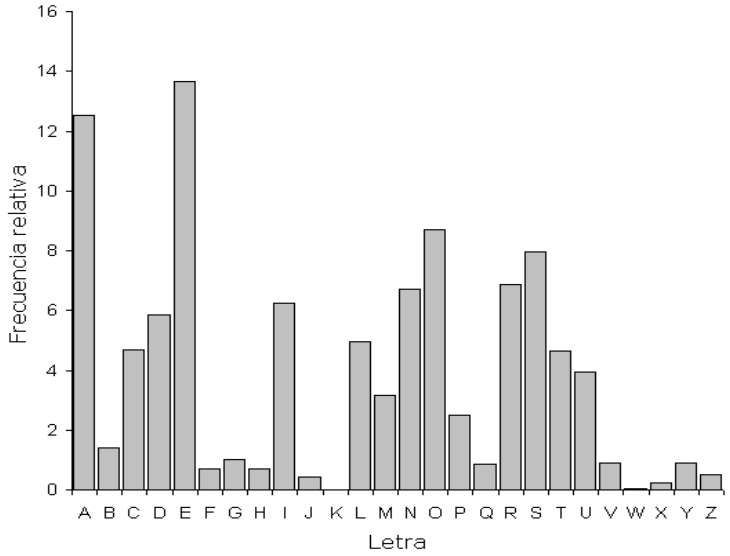
Cryptanalysis of monoalphabetic ciphers

The cryptanalyst requires the language statistics:

English



Spanish



Cryptanalysis of monoalphabetic ciphers

- Assumes that the text is long enough and typical
- Some texts may be atypical!

"THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG"

- Pangram: a short text containing every letter of the alphabet

"As his country is torn apart by social and political anarchy, Anton Vowl, a chronic insomniac, is unaccountably found missing. Ransacking his Paris flat, a group of his faithful companions trawl through his diary for any indication -- for any faint hint -- as to his location. All that it brings to light is Vowl's liking for parody wordplay and dazzling fictional constructs, which continually risk spiralling out of control, and gradually, insidiously, a ghost from Vowl's past starts to cast its malignant shadow."

- A paragraph in English with no letter 'e'! From A Void, by Georges Perec

The problem with monoalphabetic ciphers

- If the adversary is in possession of the language statistics, and has enough text, despite the very large number of possible keys, monoalphabetic ciphers do not provide much security
  - The problem is that E always encrypts to the same character (in our case, to T), and so on…
  - That is, the cipher hides the text, but does not hide the text statistics

Vigenère cipher

- Named after Blaise Vigenère (1523 – 1596)

- **Polyalphabetic Cipher**: uses different substitutions throughout the plaintext (though in regular periods)

Encryption under the Vigenère cipher

- Choose a keyword:
  - **FORCE**

Encryption under the Vigenère cipher

- Choose a keyword:
  - **FORCE**

Plaintext	T	U	R	I	N	G	W	O	R	K	E	D	I	N	I	S	O	L	A	T	I	O	N

Encryption under the Vigenère cipher

- Choose a keyword:
  - **FORCE**

Plaintext	T	U	R	I	N	G	W	O	R	K	E	D	I	N	I	S	O	L	A	T	I	O	N
Key	F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R

Encryption under the Vigenère cipher

- Choose a keyword:
  - **FORCE**

Plaintext	T	U	R	I	N	G	W	O	R	K	E	D	I	N	I	S	O	L	A	T	I	O	N
Key	F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R
Ciphertext	Y	I	I	K	R	L	K	F	T	O	J	R	Z	P	M	X	C	C	C	X	N	C	E

- Note that characters are encrypted in different ways
  - Frequency analysis is not straightforward anymore…

Cryptanalysis of Vigenère cipher

- Note that at every 5 steps, characters are encrypted using the same operation
  - The 1st, 6th, 11th, etc, characters are encrypted with **F**

T	U	R	I	N	G	W	O	R	K	E	D	I	N	I	S	O	L	A	T	I	O	N
F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R	C	E	F	O	R
Y	I	I	K	R	L	K	F	T	O	J	R	Z	P	M	X	C	C	C	X	N	C	E

- If we know the length of the key, and the text is long enough, we can treat subsets of plaintext characters as encrypted using a monoalphabetic cipher, and apply frequency analysis

Summary: cryptanalysis of Vigenère cipher

Two steps:

1. Recover the length of the keyword;
2. Use frequency analysis on sets of characters encrypted under the same key (i.e., character of the keyword)

Exercise:

- You know how to do step 2
- Do some research on how one can do step 1
  (see “week 2 Further Reading & Research” in Moodle)